Tourism Pressure and Housing Markets in Barcelona: A Counterfactual Analysis of HUT License Removal

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Abstract

This study examines how the removal of licenses for Housing Units for Tourists (HUTs) would affect Barcelona's housing market, focusing on rental and selling prices across neighborhoods. The analysis integrates spatial datasets, socioeconomic indicators, and historical housing price records to construct a counterfactual scenario in which all HUTs are eliminated. The methodological approach combines neighborhood-level harmonization, K-means clustering, spatial visualization, and regression-based impact estimation. The results reveal a clear asymmetry: rental prices tend to rise after HUT removal, while selling prices tend to fall, particularly in high-tourism areas. These effects reflect structural heterogeneity across neighborhoods and align with urban economic theory on tourism-driven housing dynamics. The findings contribute to evidence-based policymaking by quantifying the differentiated effects of tourism regulation on local housing markets.

Introduction

Barcelona has experienced sustained tourism pressure, driven in part by the proliferation of Housing Units for Tourists. These units have been linked to rising housing prices, neighborhood transformation, and tensions between tourism activity and residential needs. In response, the municipality announced the cancellation of all HUT licenses, prompting the central question of this study: **What would happen to housing prices if all HUTs were removed from the city?**

This research is relevant to urban policy debates, housing affordability concerns, and the design of tourism regulations grounded in empirical evidence.

Methodology

The study integrates multiple datasets from Open Data BCN and municipal sources, including the HUT registry, hotel locations, rental and selling price series from 2014 to 2025, population and foreign population indicators, per capita income, and neighborhood shapefiles. All datasets were standardized using a unified neighborhood identifier (neigh_norm) to ensure cross-dataset consistency.

The harmonization process involved Unicode cleaning, removal of accents and stop words, dictionary-based corrections, and consistency checks across sources. Spatial analysis included the creation of a map of HUT density, a map of K-means clusters, and spatial overlays to support interpretation.

K-means clustering (k = 3) was applied to identify three neighborhood profiles: high-income residential areas, touristic and central zones, and urban mixed areas. Counterfactual modeling was then used to estimate baseline predicted prices (with observed HUT density) and counterfactual predicted prices (with HUT density set to zero), allowing the calculation of percentage impacts on rental and selling markets.

The statistical analysis combined linear regression, LOESS smoothing, and kernel density estimation (KDE) to capture both linear and nonlinear patterns in the data.

Results

The figures included in the report serve to contextualize and visually reinforce the analytical findings.
Figure 1 illustrates the spatial distribution of HUT density across neighborhoods, highlighting areas with concentrated tourism activity.

Figure 1. HUT Density Across Neighborhoods

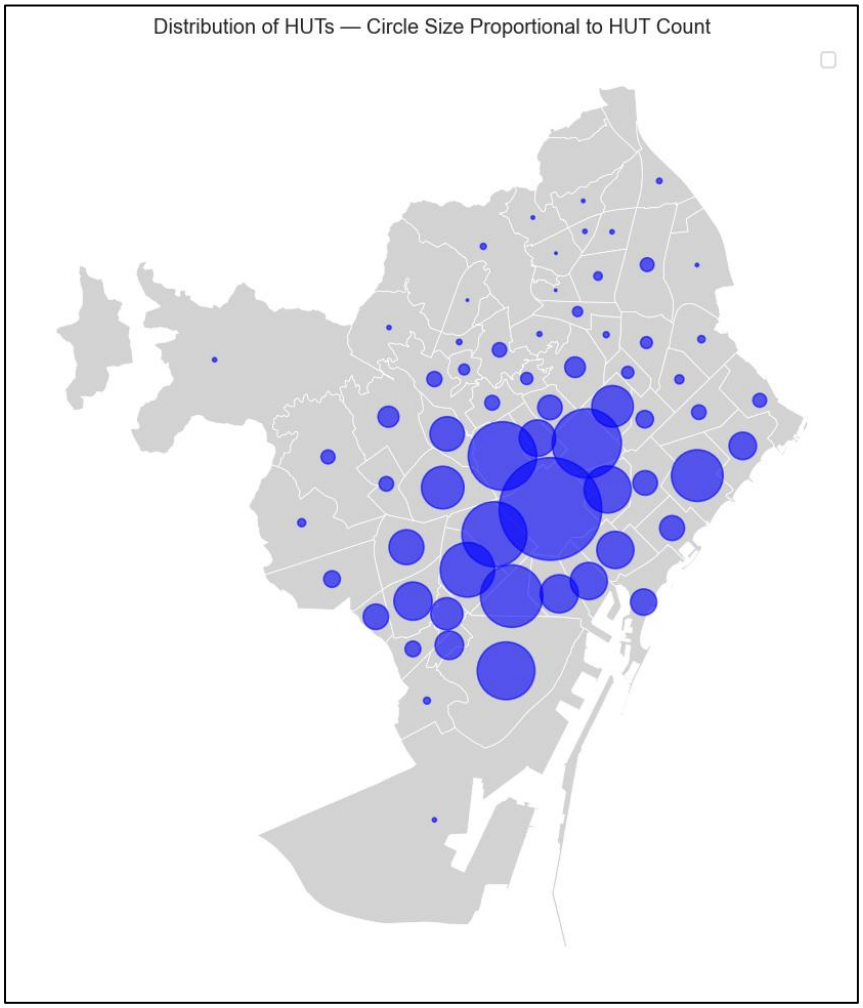


Figure 2 presents the distribution of estimated price impacts after HUT removal, showing how rental and selling markets respond differently.

Figure 2. Distribution of Price Impacts After HUT Removal

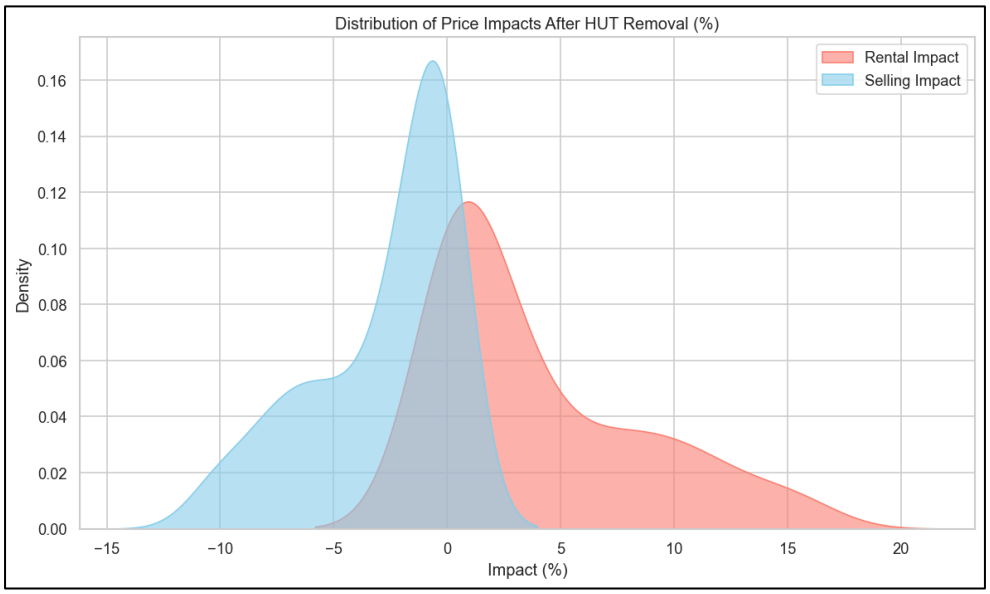
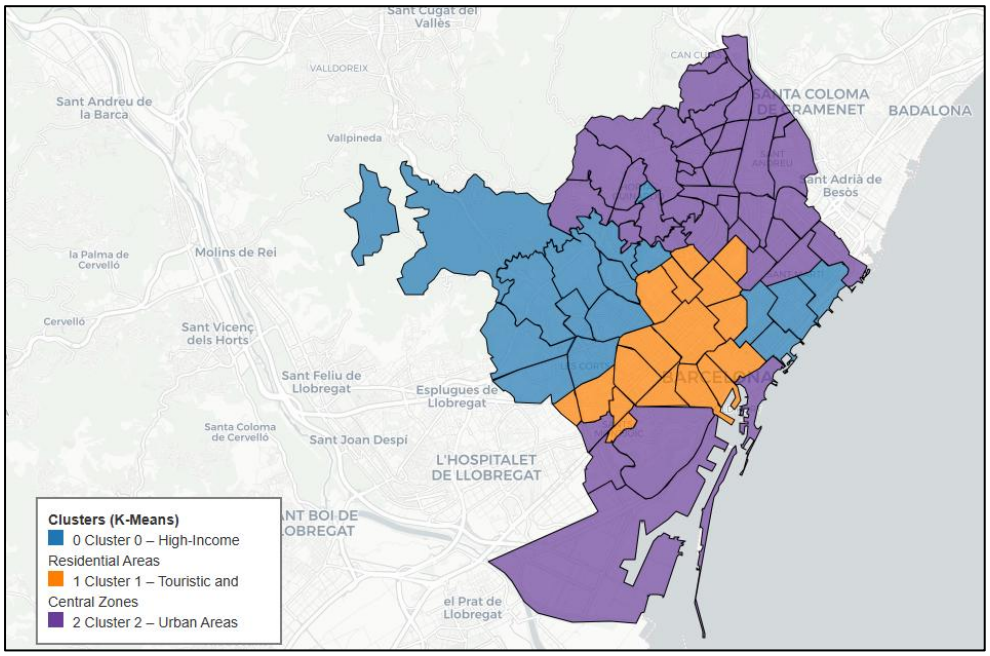


Figure 3 displays the neighborhood clusters generated by the K-means algorithm, offering a spatial interpretation of socioeconomic and tourism-related segmentation.

Figure 3. Neighborhood Clusters (K-means)



Price Impact Distribution

The distribution of estimated impacts shows that rental price effects cluster above zero, with a peak around +2%, while selling price effects cluster below zero, with a peak around –2%. This asymmetry suggests that removing HUTs reduces investment attractiveness—lowering selling prices—while simultaneously increasing pressure on the long-term rental market.

Spatial Patterns

Touristic neighborhoods exhibit both higher HUT density and stronger price impacts. The spatial patterns indicate that central and tourism-intensive areas are more sensitive to the removal of HUTs, reinforcing the idea that tourism activity is unevenly distributed across the city.

Clustering

The K-means segmentation reveals three distinct neighborhood profiles. The touristic-central cluster shows the strongest impacts, reflecting its dependence on tourism-related housing dynamics. High-income residential areas show moderate effects, while urban mixed areas display smaller impacts, consistent with their more diversified housing and economic structures.

Discussion

The findings indicate that removing HUTs produces asymmetric effects on Barcelona's housing market. Rental prices increase because units previously used for tourism re-enter the long-term market, raising demand in central areas. Selling prices decrease due to reduced investment attractiveness and lower expected returns from tourist rentals.

These results align with previous research on tourism-driven housing dynamics, which highlight the role of short-term rentals in inflating property values and reshaping urban centers. The spatial heterogeneity observed in the analysis underscores the need for geographically targeted policies rather than uniform citywide measures.

Conclusion

This study provides empirical evidence that the removal of HUTs affects rental and selling prices differently across Barcelona's neighborhoods. The counterfactual analysis demonstrates that tourism regulation has significant and spatially uneven consequences, particularly in central and tourism-intensive areas.

The research also presents several limitations. The use of annual data limits temporal precision, potential omitted variables may influence the results, and the counterfactual model relies on assumptions that may not fully capture real-world dynamics.

Future research could incorporate dynamic modelling, resident surveys, or alternative regulatory scenarios to deepen understanding of how tourism policies shape urban housing markets.

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